



SYSTEMS INTEGRATION AND ARCHITECTURE 2

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PharmaSync

By PharmaSync

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CLIENT BACKGROUND

HopeMed Pharmacy is a local pharmacy providing prescription medicines, vitamins, and healthcare supplies under pharmacist and store manager Ms. Jennifer F. Cano. Daily operations currently face bottlenecks due to manual, paper-based processes for tracking inventory, monitoring stock, and recording sales, which are highly time-consuming and prone to error. To resolve these challenges, the proposed Pharmacy Management System will replace manual paperwork with a centralized web platform that automates inventory control, streamlines transactions via a Point-of-Sale (POS) interface, and maintains real-time database synchronization. This digital transition is designed to eliminate record-keeping discrepancies, reduce administrative workloads for staff, and ensure that essential healthcare products remain reliably and promptly available to the community.

EXISTING SYSTEM INTEGRATION OVERVIEW

1. Current State vs. Proposed System

Currently, HopeMed Pharmacy operates using manual, paper-based records or standalone spreadsheets for inventory tracking, counter sales, and stock monitoring. The new Pharmacy Management System (PMS) replaces these disconnected manual workflows with a single, centralized digital platform.

2. Integration Touchpoints & Data Flow

Area	Current Manual Method	Integration & Digital Transition
Point-of-Sale (POS)	Paper receipts/logbooks for totals	Sales recording, auto-calculated totals, and auto inventory deduction all in real time.
Inventory & Stock Records	Manual shelf counts, paper/basic sheets	Migrated to central database; stock auto-updates per transaction
Supplier Communication	Manual stock checks, manual reorder lists	Low-stock alerts and exportable reports for suppliers
Administrative Reporting	Manual consolidation of sales/stock data	Auto-generated real-time sales, inventory, and audit reports

3. Initial Data Migration (Setup Step)

- **Data Cleansing & Entry:** Current product lists, medicine names, unit prices, batch numbers, expiration dates, and existing stock counts from physical ledgers will be verified and uploaded to initialize the new database.
- **Master Records Setup:** Standard user accounts (Cashier, Pharmacist, Administrator) will be configured with role-based access to secure daily transactions and administrative functions.

4. External Interfaces

- **Hardware Compatibility:** The system connects with standard counter hardware, including receipt printers and cash drawers.
- **Report Exports:** Built-in support to export reports into standard formats (PDF, CSV/Excel) for external reporting and supplier purchase orders.

SYSTEM INTEGRATION PROPOSAL

This proposal lays out how HopeMed Pharmacy's Pharmacy Management System (PMS) will be integrated, tying together the modules already inside the system and connecting them to the outside parties the pharmacy deals with every day, such as suppliers and payment channels.

1. Internal Integration (Within the PMS)

The PMS is built around one shared MySQL database, so its modules already work as one connected system rather than separate tools. This internal integration includes:

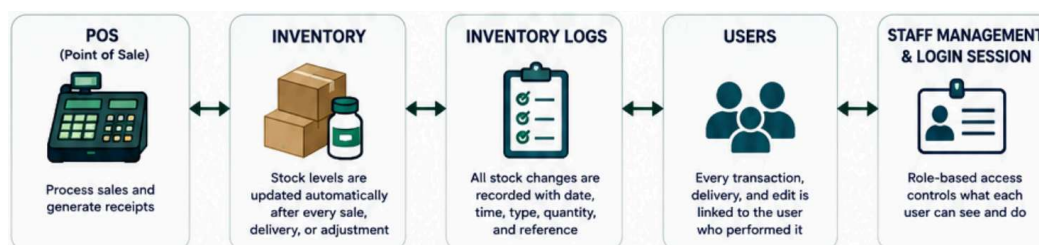


Figure 1.0 Internal Integration of HopeMed Pharmacy's Pharmacy Management System (PMS)

- **POS ↔ Inventory**: POS is linked to Inventory, so every sale automatically deducts the sold quantity from stock, with no separate step to update inventory.
- **Inventory ↔ Inventory Logs**: Every stock change, whether from a new delivery, a sale, or a manual adjustment, is automatically written to the Inventory Logs table, giving the pharmacy a full, timestamped trail of its stock movement.
- **Users ↔ Sales/Logs**: Every transaction, delivery, and inventory adjustment is linked to the responsible user account for full administrative traceability.
- **Staff Management ↔ Login Session**: Active sessions enforce role-based access control, restricting system views and permissions based on user roles.

2. External Integration (Connecting HopeMed to Outside Systems)

This is the core of the proposal, extending the PMS beyond the pharmacy's own database so it can exchange information with the parties HopeMed regularly works with.

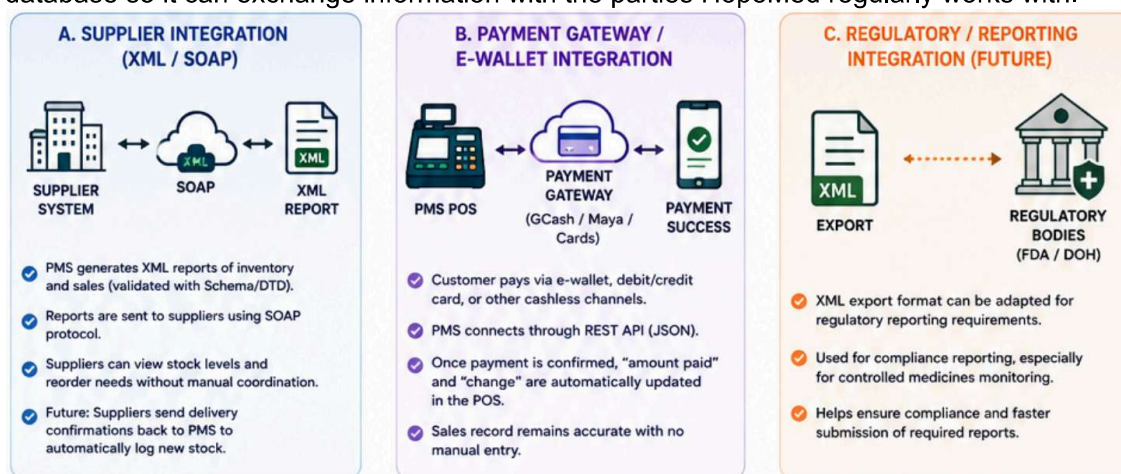


Figure 1.1 External Integration of HopeMed Pharmacy's Pharmacy Management System (PMS)

a. Supplier Integration (XML / SOAP)

- **Automated stock & sales reporting:** The PMS generates structured XML reports of inventory and sales, validated against a Schema/DTD, so the data format stays consistent and error-free every time it's sent out.
- **Reduced manual coordination:** These XML reports let suppliers see HopeMed's current stock levels and reorder needs without the pharmacy staff having to call, email, or manually prepare a separate report.
- **SOAP as the communication protocol:** SOAP is proposed as the messaging protocol for this exchange, allowing HopeMed's system and a supplier's system to communicate in a standardized, machine-readable way.
- **Future two-way exchange:** As a next step, suppliers could send delivery confirmations back into the PMS through the same channel, automatically logging new stock instead of requiring manual encoding on HopeMed's end.

b. Payment Gateway / E-Wallet Integration

- **Cashless transactions:** A REST API connection to payment processors (e.g., GCash, Maya, debit/credit card networks) is proposed so customers can pay cashless directly at the point of sale.
- **Automatic payment confirmation:** Once payment is confirmed by the gateway, the “amount paid” and “change” fields in the POS module would update automatically, keeping the sales record accurate without manual entry.

c. Regulatory / Reporting Integration (Future Scope)

- **Compliance reporting:** The same XML export structure already built for suppliers can later be adapted to meet reporting formats required by regulatory bodies such as the FDA or DOH, particularly for monitoring controlled medicines.

Integration Flow

At a high level, data is expected to move through the system in the following order:

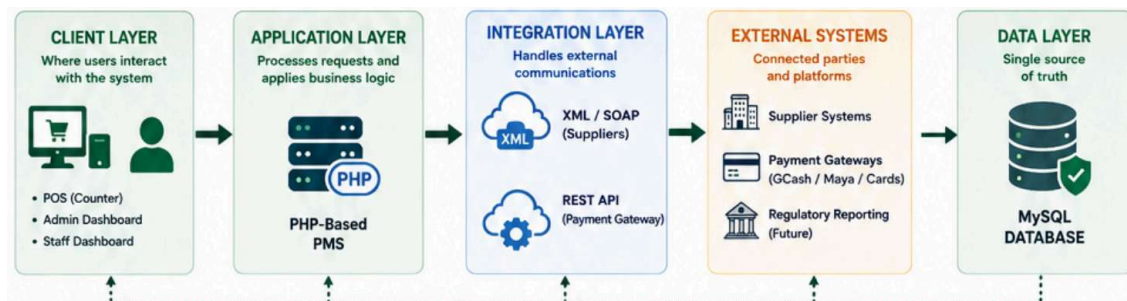


Figure 1.2 External Integration of HopeMed Pharmacy's Pharmacy Management System (PMS)

- **Client Layer:** Front-end (POS, Admin Dashboard, Staff Dashboard) — where staff and admins interact with the system.
- **Application Layer:** The PHP-based PMS that processes requests, applies business logic, and updates the MySQL database.

- **Integration Layer:** A dedicated module handling XML/SOAP generation for suppliers and REST API calls for payment gateways — this is the new layer this proposal introduces.
- **External Systems:** Supplier systems, payment gateways, and (eventually) regulatory reporting platforms.
- **Data Layer:** The single MySQL database remains the source of truth behind all of the above, so internal records and outgoing reports always match.

3. Technologies and Standards to Be Used

- PHP and MySQL - the existing core of the PMS, kept as the foundation.
- XML with Schema/DTD validation - for structured, error-checked supplier and regulatory reports.
- SOAP - as the protocol for supplier-facing data exchange.
- REST API with JSON - for lighter, faster communication with payment gateways and any future mobile or web-based extensions.
- HTTPS - as the transport layer for all data leaving or entering the system.

4. Security Measures for Integration

- **Encrypted transmission:** All data exchanged with suppliers or payment gateways will travel over HTTPS/TLS, so information cannot be intercepted in transit.
- **Authenticated access:** External systems connecting to the PMS (suppliers, payment gateways) will need an API key or token, so only authorized parties can send or pull data.
- **Consistent internal safeguards:** The hashed passwords and prepared statements already used inside the PMS will carry over to the integration layer, keeping the same protection against credential leaks and SQL injection.
- **Role-based permissions maintained:** Even through the integration layer, only the roles authorized within the PMS (e.g., admin) will be able to trigger exports or approve external data exchanges.